

Inference about means

Slide route: Lecture2v2.tex: hypothesis testing through California Test Score Data Set. Stock and Watson Chapter 3.

Session 3 | Evidence against a claim

Outcomes: State a null and alternative, compute a test statistic and confidence interval, distinguish statistical from economic significance, and compare two independent groups.

Opening A sample mean is 104; a policy report claims the population mean is 100. Is four units a large discrepancy? We need the scale of sampling uncertainty.

100-minute route

0–10	Recall sampling distributions and SE.
10–30	Hypothesis Testing, Key Concepts, Computing the P-Value.
30–45	Sample variance, standard errors, t-statistic; worked test.
45–55	Type I/II errors and short pause.
55–70	Confidence intervals and test/interval correspondence.
70–90	Mean Difference Testing, randomization, California application.
90–100	Exit exercise and preview of a regression slope.

Board example | A mean

Use with slides Lecture2v2 slides 5–11: What is Hypothesis Testing? through the t-statistic. Start the numerical mean test on slide 8, compute the standard error on slide 10, and finish the test statistic on slide 11.

Illustrative sample: $n = 100$, $\bar{y} = 104$, $s = 20$. Test $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$.

$$SE(\bar{Y}) = 20/\sqrt{100} = 2, \quad t = (104 - 100)/2 = 2.$$

Using a large-sample normal approximation, the two-sided p -value is about .0455 and a 95% interval is

$$104 \pm 1.96(2) = [100.08, 107.92].$$

Reject at 5% but not at 1%. If observations are i.i.d. normal and variance is estimated, an exact t_{99} reference gives a slightly larger p -value (about .048). State which reference distribution is being used.

What to say A p -value is the probability, under the null and assumptions, of a statistic at least as extreme as the observed one. It is not the probability that the null is true.

If short Keep the mean example and two-group example. Use one California output table deeply; assign repetitive descriptive tables for review.

Side example | Comparing groups

Use with slides Lecture2v2 slides 15–19: Confidence Intervals, Mean Difference Testing, and Causal Inference in Randomized Control Trials. Use the two-group calculation on slides 17–18, then discuss the random-assignment caveat on slide 19.

Two independently sampled groups have means 70 and 65, SDs 10 and 12, and sample sizes 50 each:

$$\hat{\Delta} = 5, \quad SE(\hat{\Delta}) = \sqrt{10^2/50 + 12^2/50} = \sqrt{4.88} \simeq 2.209.$$

$$t \simeq 2.264, \quad 95\% \text{ large-sample CI: } 5 \pm 1.96(2.209) = [.67, 9.33].$$

Ask students what changes if these are the same individuals measured twice. The independent-groups formula is then inappropriate; the covariance between measurements matters.

Causal interpretation If groups are assigned randomly and the outcome comparison follows that assignment, a mean difference can estimate a treatment effect under the design's assumptions. If students choose their own school, the same arithmetic does not remove selection.

Type I/II prompt A Type I error rejects a true null. A Type II error fails to reject a false null. Holding sample size and effect fixed, demanding a smaller significance level generally reduces power.

Confidence-interval language The method covers the fixed population parameter in 95% of repeated samples under its assumptions. Avoid saying that 95% of observations lie in this interval.

Economic significance An enormous sample can make a tiny effect statistically significant. Ask whether the interval excludes effects small enough to be irrelevant to the decision.

Exit exercise and answers

If $n = 400$ with the same SD and the same observed gap of four units, $SE = 1$ and $t = 4$. The standard error halves, not quarters.

If a 95% interval includes zero, fail to reject a two-sided zero-effect null at the corresponding 5% level. This is not proof of zero effect.

Next preparation Read Chapter 4. Bring a scatterplot of test scores against class size to the next class and ask how to summarize its slope.

Source: Lecture2v2.tex. The numerical data above are synthetic teaching examples.